

What Models Don't Know

MATHEMATICAL MODELS CAN help us make sense of the world. They depict versions of just about any area you can think of, from real estate sales to Covid deaths. But if you follow any modeled data series, you must always keep in mind statistician George E. P. Box's warning: "All models are wrong, but some are useful."

That is if your assumptions are valid and that we are not feeding bad data into them.

Most models have a fundamental flaw: They are based on the idea that the future will look a lot like the past. Nothing throws a model off more than when an event we failed to anticipate destroys its fundamental premise. Box's admonition serves as a reminder that models create a false shadow of reality. The universe is more complicated than what our mathematics suggest.

Yet, we place a great deal of faith in models. The problem occurs when, as journalist Jonathan V. Last¹⁵⁰ observes, we forget that models are "not a report sent back from the future." He observes the three types of inputs that go into models:

1. Stuff we know;
2. Stuff we think we know;
3. Stuff we have no idea about.

Sometimes, the errors that models produce are not that important. Netflix has built its business modeling what you are likely to enjoy streaming, based on your viewing habits relative to what the rest of its 167 million subscribers enjoy. If the algorithms get it wrong, the downside is the service suggests a movie you end up not liking.

Where modeling errors become more costly is when we forget they are